

Project Design Document

“Ferr-y Dust”

1. Design Intent

For context, most biometric devices (like smartwatches) present data through static screens and numbers, which are difficult to interpret during high-activity tasks or for users with visual impairments.

Our intent with this project was to explore Non-Visual Biometrics. By mapping heart rate to a pentatonic scale and a high-contrast light pulse, we create an "eyes-free" feedback loop.

2. Inspiration & References

Inspiration: The project idea came from an interest in exploring unexpected flows, specifically the idea of a biological rhythm controlling physical matter. While the initial concept used ferrofluids, the final iteration focuses on the mathematical translation of a pulse into a tonal scale. We also drew inspiration from a design studio that explores similar flows surrounding touch, called Can Touch This Studio.
<https://designwanted.com/can-touch-this-playtronica-interaction-design/>
https://cantouchthis.studio/?utm_source=ig&utm_medium=social&utm_content=link_in_bio&fbclid=PAZXh0bgNhZW0CMTEAc3J0YwZhcHBfaWQMMjU2MjgxMDQwNTU4AAAGnpnk-tDrW7W-3sxo0qphhWfaQoSXsuoJHMiHXpgDw-wpwdY01zQxBx6ORexY_aem_sngVeY6ONiYh37mHhLdQ2A

Another source of inspiration for one of our trials were Cymatics (Hans Jenny & Ernst Chladni): We looked at how physical matter reacts to frequency. We initially tried to replicate this using a Chladni plate (metal plate with sugar) to see if a heartbeat could form geometric shapes. We learned that while high frequencies form cool patterns, it was difficult to regulate and maintain one frequency. And we had to do a lot of trial and error sweeps to figure out a frequency where there was actual vibration in the plate.

3. User Journey

Approach: The user sees a 16-bit NeoPixel ring glowing in a low, "breathing" blue standby mode.

Interaction: The user places their finger on the MAX30102 sensor.

Calibration: For the first 2-3 seconds, the LEDs remain blue while the software builds a rolling average of the user's pulse.

Feedback: Once a pulse is locked, the system begins a synchronized "spike." Every time the heart beats, the NeoPixels flash bright white-blue and the speaker plays a note from a pentatonic scale.

Completion: When the finger is removed, the system kills all sound and light immediately, resetting the buffer for the next user.

4. Success Criteria

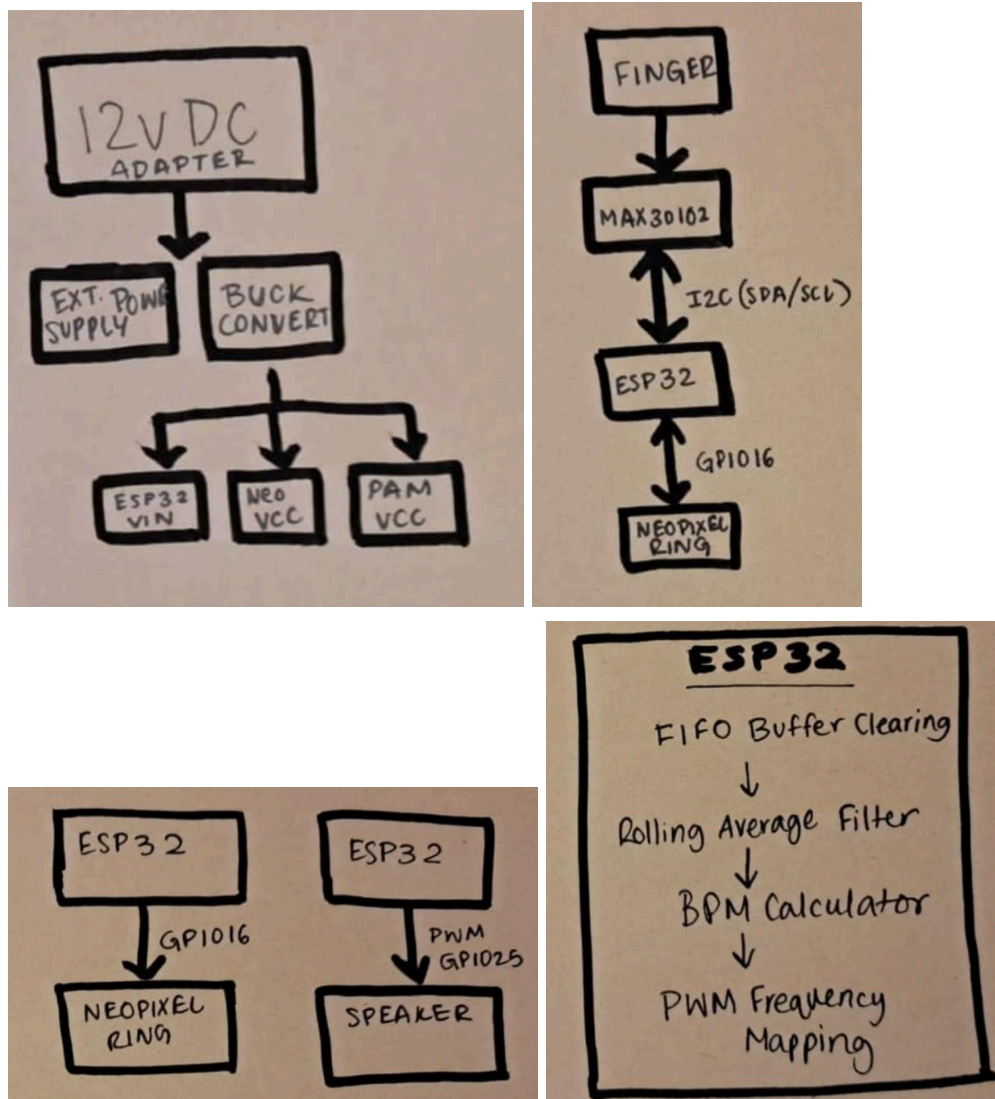
1. Speed (Response Time): The light and sound should happen at the exact same time as the heartbeat. There shouldn't be a noticeable delay that ruins the "real-time" feel.
2. System Stability: The code shouldn't crash or start printing "garbage" characters (serial noise) even if the sensor is used for several minutes straight.
3. Visual Clarity: The NeoPixels must be bright enough to be seen clearly in a well-lit classroom, and the color shift (from the idle pulse to the heartbeat spike) should be obvious to the user.
4. User Engagement: The interface should be "plug-and-play." A random person should be able to walk up, put their finger on the sensor, and understand immediately that they are controlling the sound and light.
5. Portability & Power: The system should run reliably off a single power outlet using the buck converter setup, without needing to be plugged into a laptop to stay "alive."

5. System Overview

The system is built around an ESP32 Microcontroller. It uses a MAX30102 High-Sensitivity Pulse Sensor as the input. The output is dual-channel: a 16-bit NeoPixel Ring for visual data and a Magnetic Speaker (driven by PWM) for audio data. A PAM8403 Mini-Amplifier Module that takes the Pulse Width Modulation

(PWM) signal from the ESP32 and amplifies it to drive a Magnetic Speaker. This is necessary because the ESP32 pins cannot provide enough current to move the speaker cone on their own. A Buck Converter is used to step down 12V power to 5V to ensure the heavy power draw from the LEDs doesn't "brown out" the processor. Other components that we tested with this project were 12v lifting electromagnet, TIP122 transistor, and IN4007 diode.

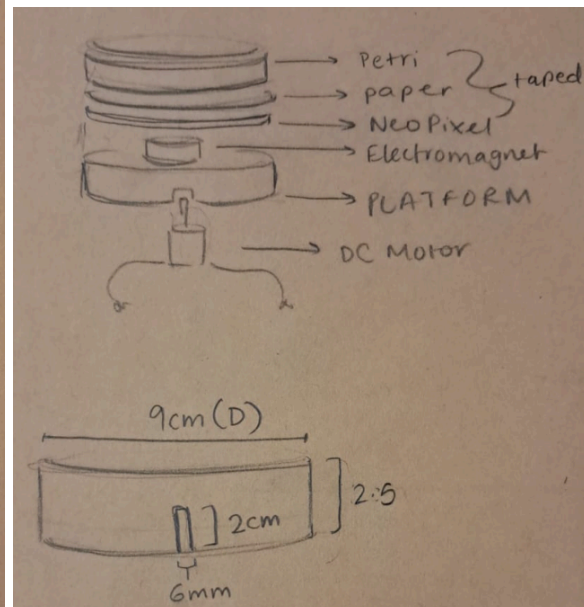
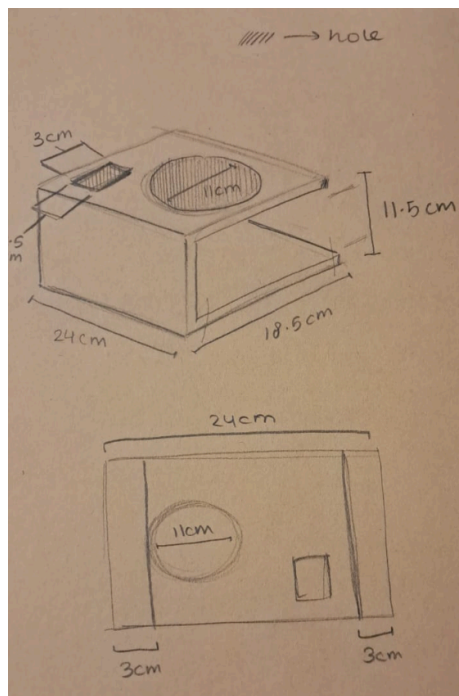
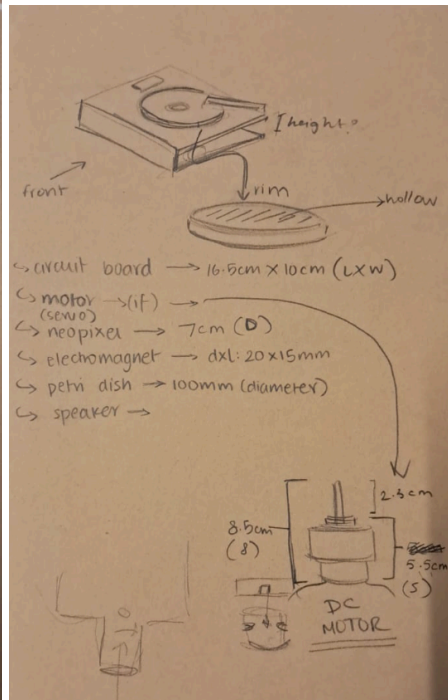
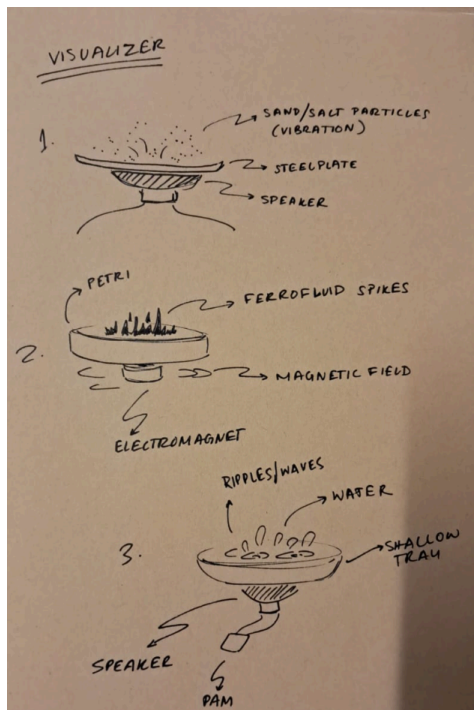
6. High-Level Block Diagram



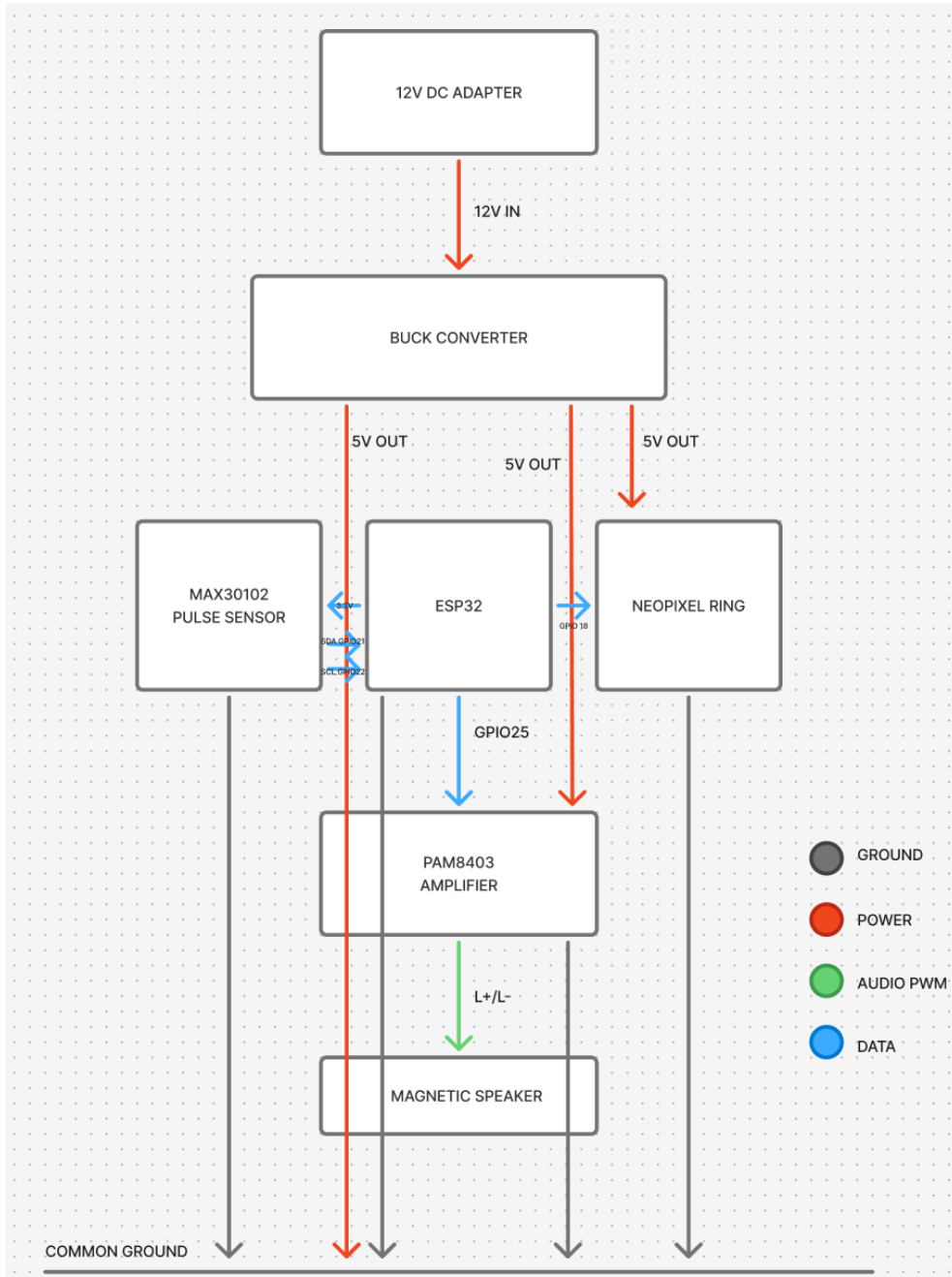
7. Input / Output Mapping

Input	Process	Output
Infrared (IR) Light: Reflected off the finger's blood flow.	FIFO Buffer Management: Clearing the sensor's memory registers.	NeoPixel Ring: Blue pulses mapped to heartbeat timing.
I2C Signal: Raw numerical data sent to the ESP32.	Rolling Average: 50-sample buffer used to find the "peak" of the beat.	PWM Audio: High-frequency tones from the pentatonic scale.
Common Ground: Ensuring all signals share a zero-volt reference.	Frequency Mapping: Converting BPM into mathematical musical notes.	Plotter Data: Real-time wave visualization in the Thonny shell.

8. Sketches & Concepts



9. Final System Architecture (Labelled)



(NOT AI.)

(Made on Figma)